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Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI&ML)

Major Project Submission (A.Y.2026-2027)

1. Project Title:

Automated Curriculum Desing using LLMs.

2. Team Details:

Team Number – 32

Team Members:

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3. Guide Name & Designation:

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4. Problem Statement:

Designing an effective curriculum is a complex and time-consuming process that requires educators to manually define course objectives, learning outcomes, module-wise content and learning resources while ensuring alignment with academic standards and industry requirements. As technologies and skill demands evolve rapidly, existing curricula often become outdated, making frequent revisions necessary. Manual curriculum development also leads to inconsistencies, increased workload, and limited personalization for different learner levels. Therefore, there is a need for an intelligent, automated solution that can efficiently generate well-structured, customizable, and up-to-date curricula. The proposed project, "**Automated Curriculum Design Using LLMs**," addresses this challenge by leveraging Large Language Models (LLMs) to automatically create comprehensive curricula based on user-defined inputs, thereby reducing effort, saving time, and improving the quality and adaptability of curriculum design.

5. Project Objectives:

Objective 1: To develop an AI-based system that automatically generates structured curricula using Large Language Models (LLMs) based on user inputs such as course title, academic level, duration, and learning objectives.

Objective 2: To generate comprehensive curriculum components, including course outcomes, module-wise topics, weekly schedules, project ideas, and recommended learning resources.

Objective 3: To reduce the time, effort, and inconsistencies involved in manual curriculum design by providing a fast, accurate, and standardized curriculum generation process.

Objective 4: To provide a customizable and adaptable curriculum framework that allows educators to review, modify, and align the generated curriculum with institutional requirements and evolving industry standards.

6. Abstract:

Curriculum design is a crucial yet time-consuming process that requires educators to develop course objectives, learning outcomes, instructional content and learning resources while ensuring alignment with academic standards and industry requirements. This project, **"Automated Curriculum Design Using LLMs,"** proposes an intelligent system that leverages Large Language Models (LLMs) to automate the creation of structured and comprehensive curricula based on user inputs such as course title, academic level, duration, and learning objectives. The system generates module-wise content, weekly learning plans, project ideas, and recommended resources, significantly reducing manual effort and ensuring consistency and adaptability. Educators can further review and customize the generated curriculum to meet institutional requirements and learner needs. By integrating artificial intelligence into curriculum planning, the proposed system enhances efficiency, supports personalized learning, and enables the rapid development of high-quality, up-to-date educational programs for schools, colleges, universities, and online learning platforms.

Keywords: Large Language Models (LLMs), Curriculum Design, Artificial Intelligence, Educational Technology, Automation, Personalized Learning.

7. Tools and Technologies Used:

Frontend: HTML, CSS, JavaScript, React (or simple web UI).

Backend: Python, Flask or FastAPI, REST APIs.

Database: SQLite / PostgreSQL for storing curriculum, user data, and performance.

LLM integration: OpenAI API, or open-source LLMs (e.g., LLaMA, Mistral) via Hugging Face.

NLP & content processing: LangChain, NLTK, or spaCy for text handling and structured outputs.

Deployment: Docker, basic cloud hosting or local server.

Optional: Mermaid/Graphviz for visualizing curriculum flow, and simple analytics dashboards.

8. Expected Deliverables:

1. **AI-Based Curriculum Generation System:** A web application that automatically generates structured curricula, including course objectives, module-wise content, learning outcomes and project ideas using Large Language Models (LLMs).
2. **Customizable Curriculum Output:** A feature that enables educators to review, edit, and customize the AI-generated curriculum according to institutional requirements, academic standards, and learner needs.

9. SDG Mapping:

SDG No	Goal Title	Justification
4	Quality Education	The project improves education by generating structured, up-to-date, and personalized curricula efficiently.
9	Industry, Innovation and Infrastructure	The project uses AI and LLMs to introduce innovation in curriculum design and educational planning.
10	Reduced Inequalities	The project provides standardized and adaptable curricula for learners with different educational needs.
17	Partnerships	The project supports collaboration between educators, institutions, and AI technologies to improve curriculum development.

10. OBE Mapping:

Program Outcomes (POs)

PO Number	Name of the PO Targeted	Justification
PO1	Engineering Knowledge	The project applies AI, LLMs, and web development knowledge to automate curriculum design.
PO2	Problem Analysis	The project identifies the limitations of manual curriculum design and provides an AI-based solution.
PO3	Design/Development of Solutions	The project develops an intelligent web application for automated curriculum generation.
PO5	Engineering Tool Usage	Use LLMs, APIs, and web frameworks.
PO7	Ethics	The project promotes the responsible and ethical use of AI in education.
PO8	Individual and Collaborative Team Work	The project encourages teamwork and effective collaboration during development.
PO11	Life-Long Learning	The project promotes continuous learning of AI and emerging technologies.

Program Specific Outcomes (PSOs):

PSO Number	Name of the PSO Targeted	Justification
PSO1	Apply the knowledge of Artificial Intelligence to design, develop, and evaluate computational solutions for complex problems in diverse domains, such as healthcare, finance, and automation	The project applies Artificial Intelligence and Large Language Models (LLMs) to automate curriculum design and provide an intelligent solution for educational planning.
PSO2	Demonstrate expertise in using advanced ML tools, techniques, and frameworks to develop innovative solutions for data analysis, pattern recognition, and intelligent decision-making system	The project uses advanced AI tools and LLM frameworks to generate structured curricula, supporting intelligent decision-making and automation.

Course Outcomes (COs):

CO1 - Apply fundamental and disciplinary concepts and methods in ways appropriate to their principal areas of study.

CO2 - Demonstrate skill and knowledge of current information and technological tools and techniques specific to the professional field of study.

CO3 - Identify, analyze, and solve problems creatively through sustained critical investigation.

CO4 - Demonstrate awareness and application of appropriate personal, societal, and professional ethical standards.

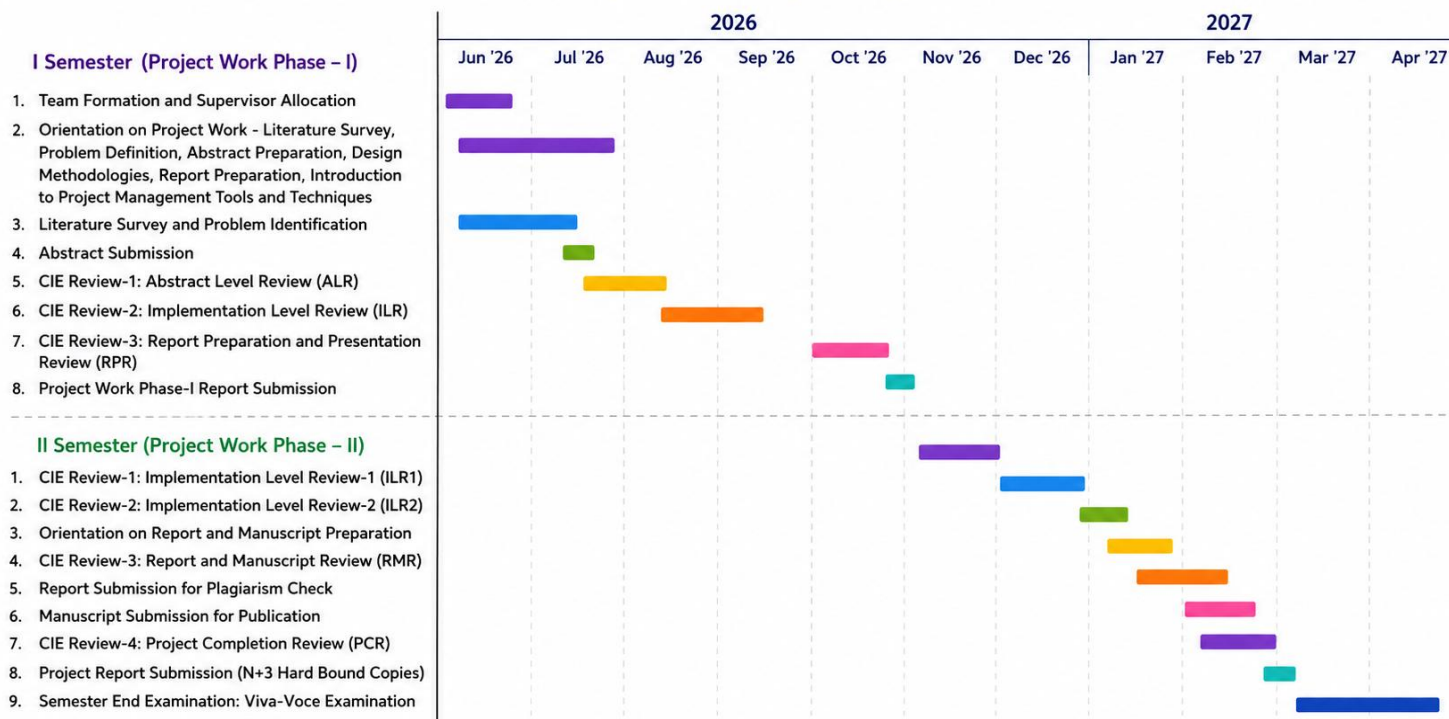
CO5 - Practice the skills, diligence, and commitment to excellence needed to engage in lifelong learning.

Bloom's Taxonomy Level:

Create (Focusing on the synthesis and design of a new curriculum structure).

11. Timeline – 2026

Project Timeline: Automated Curriculum designs using LLMs (June 2026 – April 2027)



12. References

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Signature of the Supervisor

Signature of Coordinator

Signature of HOD